The Data Refinery — 21 Day Learning Sprint

Day 2/21: OLTP vs. OLAP Paradigms & Architecture Challenges

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🧠 Core Concept Recap: OLTP vs. OLAP Storage Engines

Choosing between row-oriented operational stores and column-oriented analytical warehouses dictates pipeline cost, latency, and scalability.

⚡ OLTP (Row-Oriented)

- Fast single-row reads/writes
- High concurrency & ACID guarantees
- *Tech:* PostgreSQL, MySQL, SQL Server

📊 OLAP (Column-Oriented)

- Columnar scans & rapid aggregations
- Reads millions of rows efficiently
- *Tech:* Snowflake, BigQuery, Redshift

Challenge 1: High-Volume E-Commerce Aggregation

MAIN CHALLENGE

You are designing a data pipeline for an E-commerce platform. Which storage approach is **MOST suitable** for computing the average order value across 50 million historical records?

A) **OLTP (Row-Oriented):** Scans every full row including user addresses and order notes.

B) OLAP (Column-Oriented): Reads only the `order_amount` column from disk. [CORRECT]

C) **Key-Value Store:** Performs a bulk lookup using key `user_id`.

D) **Document DB:** Loads nested JSON documents one by one into memory.

Detailed Explanation:
Column-oriented databases (OLAP) store each column's data contiguously on disk. To compute `AVG(order_amount)` across 50M records, the engine performs *I/O pruning*, reading only the byte range containing the `order_amount` column. An OLTP database must read full horizontal rows off disk, bringing unnecessary metadata into memory and creating massive disk I/O bottlenecks.

Challenge 2: Food Delivery System Workload Pairing

SCENARIO CHALLENGE

A food delivery platform requires two distinct systems:
1. Instantly update order status when accepted by a delivery driver.
2. Calculate the top 10 most ordered dishes nationwide over the past quarter for marketing.

A) Task 1: OLAP | Task 2: OLTP

B) Task 1: OLTP | Task 2: OLAP [CORRECT]

C) Task 1: OLTP | Task 2: OLTP

D) Task 1: OLAP | Task 2: OLAP

Detailed Explanation:
• **Task 1 (OLTP):** Requires high concurrency and sub-second point updates with strict ACID compliance when drivers accept orders.
• **Task 2 (OLAP):** Requires scanning millions of historical order line items across a 90-day window to aggregate popular

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Challenge 3: Technical Mechanics — Storage Efficiency

TECHNICAL MECHANICS

Why does an **OLAP (Column-Oriented)** database like BigQuery or Snowflake drastically reduce query costs and execution time when running `SELECT AVG(sale_price) FROM sales` compared to a traditional row-oriented OLTP database?

A) It automatically caches all table data in RAM on startup.

B) It skips reading non-relevant columns (like `customer_id`, `address`, `item_description`) off disk entirely. [CORRECT]

C) It disables ACID transactions to bypass lock waits.

D) It converts SQL directly into C++ compiled binaries before reading data.

Detailed Explanation:

- **I/O Optimization & Pruning:** In columnar storage, values for each column are stored sequentially on physical storage blocks. When executing `SELECT AVG(sale_price)`, the engine performs column pruning, bypassing all other fields entirely.
- **Cost Impact:** In modern cloud data warehouses like BigQuery, billing is directly calculated based on the total gigabytes/terabytes of data scanned from disk. By skipping unreferenced columns, you reduce scanned volume by 80%–95%, proportionally dropping query execution costs and execution time.
- **Compression Efficiency:** Because columns contain uniform data types (e.g., all integers or floats), columnar files compress significantly better (using dictionary or run-length encoding) than mixed row data.

📋 Quick Answer Key Summary

Challenge	Focus Area	Correct Option
Challenge 1	E-Commerce 50M Record Aggregation	**B (OLAP Column-Oriented)**
Challenge 2	Food Delivery Workload Pairing	**B (Task 1: OLTP | Task 2: OLAP)**
Challenge 3	Column Pruning & Cost Reduction	**B (Skip non-relevant columns off disk)**

📌 *Note for group members: Keep sharing your reasoning in "The Data Refinery" WhatsApp community chat! Day 3 challenge drops tomorrow.*

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